

1). A field: ~~Abelian group~~

$$F = (A, +, \cdot),$$

where  $A$  forms an Abelian group under  $+$ ,

$A/\{0\}$  forms an Abelian group under  $\cdot$ .

for  $a, b, c \in A$

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c)$$

And ~~every~~  $\forall a \in A/\{0\}$ ,

$a$  is a unit,

i.e.  $\exists a^{-1}$  where  $aa^{-1} = e. \forall a \in A/\{0\}$



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A field has the following properties:

I) Under  $+$ ,  $(A, +)$  is an Abelian group.

- i) Closure
- ii) Associativity
- iii) Identity ( $e_+ = 0$ )
- iv) Inverse
- v) Commutative

II) Under  $\cdot$ ,  ~~$(A, \cdot)$~~   $(A/\{0\}, \cdot)$  is an abelian group.

- i) Closure
- ii) Associativity
- iii) Identity ( $e_\cdot = 1$ )
- iv) Inverse -  $\forall$  every element is a unit  $\neq 0$
- v) Commutative

III) Distributivity ( $\cdot$  over  $+$ )



Now, for  $F_2 = (C, +, \cdot)$ , where  
 $x \in C$  is of form  $a+bi$   
 Let Subfields be

a).  $F_1 = (A, +, \cdot)$   $A = \mathbb{Q} + 0i$ .

where  $x \in A$  is of the form

$$x = a + 0(i), \quad a \in \mathbb{Q}$$

So,  $A \subset C$  where  $b=0$

Now, to prove  $F_1 \subset F$

$F_1$  is a subfield of  $F$

i).  $F_1$  is NON empty.

for  $a=0$ ,

$$x = 0 + 0i, \text{ so } x \in F_1.$$

$x$  is  $e_+$ , it has an inverse

$$(x)^{-1} = -x \in F_1.$$

and it shows



$F_1$  is closed under  $+$ .

for  $x = a + 0i$  and  $y = b + 0i$ .

$$\begin{aligned} x + y &= a + 0i + b + 0i \\ &= (a + b) + 0i \end{aligned}$$

so,  ~~$a \in \mathbb{R}$  and  $y \in \mathbb{R}$~~   
 ~~$x + y \in \mathbb{R}$ .~~

$$\therefore \text{ ~~$(a + b) \in \mathbb{R}$~~ }$$

Now, as  $a \in \mathbb{Q}$ ,  $b \in \mathbb{Q}$   
 $a + b \in \mathbb{Q}$

$$\therefore (a + b) + 0i \in F_1$$

$\therefore F_1$  is closed under  $+$ .

Now, Associativity and Commutativity are shown by  $\mathbb{C}$

$\therefore$  for  $A \in \mathbb{C}$ , they will automatically hold true.

Now, as  $0 + 0i \in \mathbb{Q}$ ,  
 we have  $e_+$  (additive identity)



So, for additive inverse  
for  $\forall a + 0i$

Let  $y$  be such that

$$a + 0i + y = 0$$

$$\therefore a + 0i + y = 0 + 0i$$

$$y = -a + 0i$$

Now, as  $a + 0i \in C$ ,  
 $-a + 0i \in C$

$\therefore C$  is abelian  
group under  $+$ .

$$\therefore -a + 0i \in A.$$

iii). ~~Closure under~~  <sup>$R$  has</sup> Closure under  $\cdot$ .

for  $x = a + 0i$  &  $y = b + 0i$

$$x \cdot y = (a + 0i)(b + 0i)$$

$$= ab + 0(a)i + 0(b)i - 0$$

$$= ab$$

for  $a \in \mathbb{Q}$ ,  $b \in \mathbb{Q}$ ,  $ab \in \mathbb{Q}$ .

Now, as Associativity and Commutativity  
are true for  $C$ ,



for ACC, they're automatically true.

Now, for  $n = 1 + 0i$ ,

$e =$  multiplicative identity  $= x$ .

and  $n \in A$ .

for every element

$$n = a + 0i, \quad n \neq \{0\}$$

~~8~~. Let there be  $y$

such that  $n \cdot y = 1$

$$\therefore (a + 0i)(y) = 1$$

$$(y) = \frac{1}{a + 0i} = \frac{a - 0i}{a^2 - 0}$$

$$= \boxed{\frac{1}{a}}$$

$$\therefore \boxed{y = \frac{1}{a} + 0i}$$

for all  $a \in \mathbb{Q}$ ,  $\exists \frac{1}{a} \in \mathbb{Q}$

~~yes~~  $y \in A$ .



Hence, every element is a unit.

iv). For distributivity of  $\cdot$  over  $+$

$$\text{for } x = a + 0i$$

$$y = b + 0i$$

$$z = c + 0i$$

$$a, b, c \in \mathbb{Q}$$

$$\begin{aligned} x \cdot (y + z) &= (a + 0i) (b + 0i + c + 0i) \\ &= (a + 0i) (b + c + 0i) \end{aligned}$$

$$\begin{aligned} &= a(b + c) + 0i + 0i - 1(0) \\ &= a \cdot (b + c) \end{aligned}$$

Now, for  ~~$\mathbb{Q}$~~   $x, y, z \in \mathbb{C}$ ,  
 ~~$\mathbb{Q}$~~  distributivity holds.

$$\therefore a \cdot (b + c) = (ab + ac)$$

$$\begin{aligned} &= (a + 0i)(b + 0i) \\ &\quad + (a + 0i)(c + 0i) \end{aligned}$$

$$= (ay + az)$$

$\therefore$  Distributivity holds.

$\therefore F_1 = (A, +, \cdot)$ , where

$$A = \mathbb{Q} + 0i$$

$$A = \mathbb{Q} + 0i$$



$$F_2 = (B, +, \cdot),$$

$$B = \mathbb{Q} + (\sqrt{2})\mathbb{Q}i$$

Let  $n \in B$ .

$$\therefore n = a + \sqrt{2}bi, \text{ where}$$

$$a, b \in \mathbb{Q}.$$

1) Non empty

for  $a, b \in \mathbb{Q}$  and  $0 \in \mathbb{Q}$ ,

we can have  $a = 0$  and  $b = 0$  for some  $a, b$

$$\therefore n = 0 + (\sqrt{2})0i = 0$$

$$\therefore n = 0 \in B.$$

$\therefore F_2$  is non-empty.

2) closure under +.

$$\text{for } x = a + b\sqrt{2}i$$

$$y = c + d\sqrt{2}i$$

$$x + y = a + b\sqrt{2}i + c + d\sqrt{2}i$$



$$= (a+c) + ((b+d)\sqrt{2})i$$

now, for  $a, c \in \mathbb{Q}$  and  $b, d \in \mathbb{Q}$ ,  $b \in \mathbb{Q}$

as ~~rational~~ <sup>rational</sup> no.s are closed under +

$$= \mathbb{Q} + \mathbb{Q}\sqrt{2}i$$

$$\in B.$$

$\therefore$  As  $B \subset \mathbb{C}$ , closure, associativity hold true for  $B$  as they are true for  $\mathbb{C}$ .

Now,  $e + = 0 \in B$  (as shown by (i)),

$$\text{So, let } x = a + b\sqrt{2}i$$

$$y = c + d\sqrt{2}i$$

such that  $xy = 0$

$$\therefore a + b\sqrt{2}i \neq c + d\sqrt{2}i = 0$$

$$c + d\sqrt{2}i = -a - b\sqrt{2}i$$

$$\therefore c = -a, d = -b$$

$$\therefore -a - b\sqrt{2}i = y$$

and



Hence, ~~B~~  $B$  is an abelian group under  $+$ .

Closure under  $\times$ .

$$\text{for } x = a + b\sqrt{2}i \\ y = c + d\sqrt{2}i$$

$$x \cdot y = (a + b\sqrt{2}i) \cdot (c + d\sqrt{2}i)$$

$$= ac + cb\sqrt{2}i + ad\sqrt{2}i - 2bd$$

$$= (ac - 2bd) + (cb + ad)\sqrt{2}i$$

Now, for  $a, b, c, d \in \mathbb{Q}$   
 $ac, ad, bd, bc \in \mathbb{Q}$

$$= \mathbb{Q} + \mathbb{Q}\sqrt{2}i$$

$$\in B$$

$\therefore$  It's closed under  $\times$ .

Now, for as  $B \subset \mathbb{C}$  and  $\mathbb{C}$  has associativity & commutativity they also hold true for  $B$ .



Now, for  $x = a + b\sqrt{2}i$   
 $\int a = 1 \quad b = 0$   
 $x = 1 = e.$

$\therefore$  it has multiplicative identity

Now, for

$$x = a + b\sqrt{2}i$$

$$\exists y \text{ such that}$$

$$x \cdot y = 1$$

$$\therefore y = \frac{1}{x} = \frac{1}{a + b\sqrt{2}i}$$

$$= \frac{a - b\sqrt{2}i}{a^2 - 2b^2} \quad \text{where } a^2 - 2b^2 \neq 0$$

$$= \frac{a}{a^2 - 2b^2} - \frac{b\sqrt{2}i}{a^2 - 2b^2}$$

$$= \odot + \sqrt{2}\odot i$$

$$\therefore y \in B.$$

$\therefore$  Every element is a unit



# Distributivity

Now, as Distributivity holds  
 for  $c$ , it also holds for

$$B \in C$$

Hence  $f_2 = (B, +, \cdot)$  where

$$B = \{3 + \sqrt{2}Q\}$$

is a subfield  
 of  $F$ .



2). Let  $e_1, e_2, e_3$  be the three types of elementary row operations to be performed on a matrix  $A_{m \times n}$ .

where  $A_{m \times n} = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}_{m \times n}$

i).  $e_1 : A_{m \times n} \rightarrow A_{m \times n}$

$$e_1(A)_{ij}^0 = \begin{cases} cA_{ij}^0 & , i = r \\ A_{ij}^0 & , i \neq r \end{cases}$$

for  $r \leq m$  and  $c \neq 0$ .

~~ii).  $e_2$~~

Now, let

$e' : A_{m \times n} \rightarrow A_{m \times n}$

$$e'(A)_{ij}^{00} = \begin{cases} k c A_{ij}^0 & , i = r \\ A_{ij}^0 & , i \neq r \end{cases}$$



Now,

of  $x = r$ ,

$$e_1(e_1(A))_{ij} = \begin{cases} k(e_1(A)_{ii}^0), & i=j=r \\ e_1(A_{ij}^0), & i \neq r \end{cases}$$

$$= \begin{cases} k(A_{ii}^0), & i=r \\ A_{ij}^0, & i \neq r \end{cases}$$

$= B$  (1)

$$e_1(e_1(A))_{ij} = \begin{cases} c(e_1(A_{ii}^0)), & i=r \\ e_1(A_{ij}^0), & i \neq r \end{cases}$$

$$= \begin{cases} k(A_{ii}^0), & i=r \\ A_{ij}^0, & i \neq r \end{cases}$$

$= B$  (2)

$$\therefore e_1(e_1(A)) = e_1(e_1(A)) = B$$

such that  $B \neq A$ .



where  $B =$

$$\begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ \vdots & \vdots & & \vdots \\ k(A_{11}) & k(A_{12}) & \dots & k(A_{1n}) \\ \vdots & \vdots & & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}$$

$m \times n$

Now, for  $n \neq r$ .

$$e'(e_1(A))_{ij}^0 = \begin{cases} k(e_1(A_{ij}^0)), & i=j=n \\ e_1(A_{ij}^0), & i \neq n \end{cases}$$

$$= \begin{cases} k(A_{ij}^0), & i=n \\ e_1(A_{ij}^0), & i \neq n \\ A_{ij}^0, & i \neq n, i \neq r \end{cases}$$

$$= B'$$



$$e_i(e'(A)_{ij}) = \begin{cases} c(e'(A))_{ii}^o, & i=r \\ e'(A)_{ii}^o, & i \neq r \end{cases}$$

$$= \begin{cases} cA_{ii}^o, & i=r \\ kA_{ii}^o, & i=r \\ A_{ii}^o, & i \neq r, i \neq r \end{cases}$$

$$= B'$$

where  $B' =$

$$\begin{bmatrix} \vdots & & & \\ cAr_1 & cAr_2 & \dots & cAr_n \\ \vdots & & & \\ kAr_1 & kAr_2 & \dots & kAr_n \\ \vdots & & & \end{bmatrix}_{n \times n}$$

$$\therefore e(e'(A)) = e'(e(A)) \neq B$$

$$\text{but } B \neq A$$



Now, for  $e_2(A)_{ij}^0$ ,

$$e_2(A)_{ij}^0 = \begin{cases} A_{ij}^0 + c A_{ij}^0, & i=j=n \\ A_{ij}^0, & i \neq n \end{cases}$$

Let there be  $e'(A)_{ij}^0$  such that

$$e'(A)_{ij}^0 = \begin{cases} A_{ij}^0 + k A_{ij}^0, & i=j=n \\ A_{ij}^0, & i \neq n \end{cases}$$

Now, if  $n=n$

$$e'(e_2(A)_{ij}^0) = \begin{cases} e_2(A)_{ij}^0 + k(A_{ij}^0), & i=j=n \\ e_2(A)_{ij}^0, & i \neq n \end{cases}$$

if  $y=1$



$$= \begin{cases} A_{ij}^{\circ} + (c+ke) A_{sj}^{\circ}, & i \neq r \\ A_{ij}^{\circ} & i = r \end{cases}$$

$$= B_1^{\circ}$$

if  $y \neq s$

$$= \begin{cases} A_{ij}^{\circ} + c A_{sj}^{\circ} + k A_{ry}^{\circ} \\ A_{ij}^{\circ} \end{cases}$$

$$e(e(A)_{ij}^{\circ}) = \begin{cases} e(A_{ij}^{\circ}) + c e(A_{sj}^{\circ}), & i \neq r \\ A_{ij}^{\circ} & i = r \end{cases}$$

if  $(s=y)$

$$= \begin{cases} A_{ij}^{\circ} + c A_{sj}^{\circ} + (k+c) A_{sj}^{\circ} \\ A_{ij}^{\circ} \end{cases}$$

~~else~~,

$$= B_1$$



$\nabla s \neq y$

$$= \left\{ \begin{array}{l} A_{ij}^0 \vdash c A_{sj}^0 \vee A_{ty}^0, i \neq n, r \\ A_{ij}^0, i \neq r \end{array} \right.$$

$= B_2$

$\therefore$  for  $n \neq r$ ,

$$e'(e_2(A)_{ij}^0) = e_2(e'(A)_{ij}^0) = B$$

while  $A \neq B$

★ ~~§ either c or k~~ atleast one of  $c$  or  $k \neq 0$  otherwise  $B = A$

Now, for  $i \neq n \in r$  and  $n \neq s, y \neq r$

$$e'(e_2(A)_{ij}^0) = \left\{ \begin{array}{l} e_2(A)_{ij}^0 + e_2(A)_{ij}^0 \\ e_2(A)_{ij}^0 \\ e_2(A)_{ij}^0 \end{array} \right.$$



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$$= \begin{cases} A_{ij}^{\circ} + k A_{ij}^{\circ} & , i = r \\ A_{ij}^{\circ} + c A_{ij}^{\circ} & , i = r \\ A_{ij}^{\circ} & , i \neq r \neq r \end{cases}$$

= B\_3.

$$e_2(e'(A)_{ij}^{\circ}) = \begin{cases} e'(A)_{ij}^{\circ} + c e'(A)_{ij}^{\circ} & , i = r \\ e'(A)_{ij}^{\circ} & , i = r \\ e'(A)_{ij}^{\circ} & , i \neq r, i \neq r \end{cases}$$

$$= \begin{cases} A_{ij}^{\circ} + c A_{ij}^{\circ} & , i = r \\ A_{ij}^{\circ} + c A_{ij}^{\circ} & , i = r \\ A_{ij}^{\circ} & , i \neq r \neq r \end{cases}$$

= B\_3.



But,

for  $e_2$ ,

if we take the case where:

$$n \neq y \quad r \neq s$$

But  $n = s$ .

Then,

for

$$e'(e_2(A))_{ij}^{\circ\circ} = \begin{cases} e_2^{\circ}(A)_{ij}^{\circ} + \delta e_2(A)_{ij}^{\circ}, & i = n \\ e_2(A)_{ij}^{\circ}, & i \neq n \end{cases}$$

Now,  $e_2(A)_{nj}^{\circ} = e_2(A)_{nj}^{\circ} = A_{nj}^{\circ}$

$$= \begin{cases} A_{nj}^{\circ} + \delta A_{nj}^{\circ}, & i = n \\ A_{nj}^{\circ} + \delta [A_{nj}^{\circ} + \delta A_{nj}^{\circ}], & i = r \\ A_{ij}^{\circ}, & i \neq n, i \neq r \end{cases}$$



$$= e'(e_2(A)_{ij}^0)$$

$$= \begin{cases} A_{ij}^0 + b_1 A_{ij}^0, & i = n \\ A_{ij}^0 + c A_{ij}^0 + b_1 A_{ij}^0, & i = n \\ A_{ij}^0, & i \neq n, i \neq r \end{cases}$$

$$= B_1$$

But, for

$$e_2(e'(A)_{ij}^0)$$

$$= \begin{cases} e'(A)_{ij}^0 + c(e'(A)_{ij}^0), & i = n \\ A_{ij}^0, & i \neq n, i \neq r \end{cases}$$

$$= \begin{cases} A_{ij}^0 + c A_{ij}^0, & i = n \\ A_{ij}^0 + b_1 A_{ij}^0, & i = n \\ A_{ij}^0, & i \neq n, i \neq r \end{cases}$$

$$= B_2$$



but, for  $B_1$  and  $B_2$

at  $i = r$ ,

$$\underbrace{A_{rj} + cA_{sj}}_{\text{in } B_1} \neq \underbrace{A_{rj}}_{\text{in } B_2} \quad \text{for } A_{sj} \neq 0$$

for  $cA_{sj} \neq 0$

so, there's a case where the equality fails.

hence, there CANNOT be an elementary row operation of the same type as  $e_2$

such that for all cases,

$$e'(e_2(A)) = e_2(e'(A)) = B$$

where  $A \neq B$ .



for  $e_3(A)_{ij}^{\circ\circ}$   $e_3: A_{m \times n} \rightarrow A_{m \times n}$

$$e_3(A)_{ij}^{\circ\circ} = \begin{cases} A_{ij}^{\circ\circ} & , i = r \\ A_{rj} & , i = r \\ A_{ij} & , i \neq r, i \in M \end{cases}$$

Let  $e'_3(A)_{ij}^{\circ\circ} : e'_3: A_{m \times n} \rightarrow A_{m \times n}$

$$e'_3(A)_{ij}^{\circ\circ} = \begin{cases} A_{ij}^{\circ\circ} & , i = r \\ A_{ij}^{\circ\circ} & , i = r \\ A_{ij} & , i \neq r, i \in M \end{cases}$$



now, for  $n \geq r$ ,

$y$  has to be equal to  $s \therefore \boxed{n \geq r} \& \boxed{y = s}$

$$\therefore e'(e_3(A)_{ij}) = \begin{cases} e_3(A)_{ij}^0, & i \geq r \\ e_3(A)_{ij}^0, & i \geq y \\ e_3(A)_{ij}^0, & i \neq n, i \neq y \end{cases}$$

$$= \begin{cases} A_{ij}^0, & i \geq n \\ A_{ij}^0, & i \geq y = s \\ A_{ij}^0, & i \neq n, i \neq y \end{cases}$$

$$= B_3$$

Similarly,

$$e_3(e'(A)_{ij}^0) = \begin{cases} e'(A)_{ij}^0, & i \geq r \\ e'(A)_{ij}^0, & i \geq s \\ A_{ij}^0, & i \neq r, i \neq s \end{cases}$$

$$= \begin{cases} A_{ij}^0, & i \geq r \geq x \\ A_{ij}^0, & i \geq s \geq y \\ A_{ij}^0, & i \neq r, i \neq s \end{cases}$$

$$= B_3$$



Similarly ~~if  $y \neq s$~~  for  $y \neq s$

$$= \begin{cases} A_{ij}^{\circ}, & i \neq x \\ A_{ij}^{\circ}, & i \neq y \\ A_{ij}^{\circ}, & i \neq r \\ A_{ij}^{\circ}, & i \neq s \\ A_{ij}^{\circ}, & i \neq x, i \neq y, i \neq r, i \neq s \end{cases}$$

$= B_5.$

Now, for  $x \neq r$

$$e_3(e'(A_{ij}^{\circ})) = \begin{cases} e'(A)_{ij}^{\circ}, & i \neq r \\ e'(A)_{ij}^{\circ}, & i \neq s \\ e'(A)_{ij}^{\circ}, & i \neq s, i \neq r \end{cases}$$



$\therefore$  for  ~~$i=r$~~  and  ~~$s=y$~~

$n=r$  and  $y=s$ , where  $n \neq y$ ,

$$e'(e_3(A_{ij}^{\circ})) = e_3(e'(A_{ij}^{\circ})) = B$$

where  $A \neq B$ .

Now, for  $n \neq r$ ,  $\boxed{y \neq s}$  necessary condition.

$$e'(e_3(A_{ij}^{\circ})) = \begin{cases} e_3(A)y_i^{\circ}, & i=r \\ e_3(A)y_j^{\circ}, & i=y \\ e_3(A)_{ij}^{\circ}, & i \neq r \end{cases}$$



for  $y \neq s$ ,

$$z \left\{ \begin{array}{l} A_{ij}^0, i \geq r \\ A_{ij}^0, i \geq s \\ A_{ij}^0, i \geq n \\ A_{ij}^0, i \geq y \\ A_{ij}^0, i \neq n, i \neq y, i \neq r, \\ i \neq s. \end{array} \right.$$

$z \notin S$ .

$\therefore$  for  $y \neq n$ ,  $r \neq n$

where  $\boxed{y \neq s}$ ,



But,

for  $c_3$ , what if,

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$\forall x \neq y$

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$$y = 1$$

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$$e'(e_3(A)^{ij}) =$$

$$e_3(A_{ij}) = x_{2j}$$

$$e_3(A) z_i^0, i=2, 3$$

$$e_3(A)^{\hat{i}} \quad , \in \mathbb{N},$$

$$y = \sum e_j(A) y_j = e_3(A) y_j = A x_j \quad \text{is } y$$

$$= \int A(x) dx = x$$

$\Rightarrow A_{ij}^0, i, j = 1, 2, \dots, n$

$\text{Arg}_i^0, i=1$

$$A_i^0, \quad i \in N, \\ i \in r, \\ i \in y.$$

$$= B_1$$



But,

for  $e_2(e'(A)_{ij}^0)$

$$= \begin{cases} e'(A)_{sj}^0 & i=r \\ e'(A)_{rj}^0 & i=s=y \\ e'(A)_{ij}^0 & i \neq r, i \neq y \end{cases}$$

Now, as  $s=y$ ,

$$e'(A)_{sj}^0 = e'(A)_{yj}^0 = A_{rj}^0$$

$$= \begin{cases} A_{rj}^0 & i=r \\ A_{rj}^0 & i=s=y \\ A_{yj}^0 & i=r \\ A_{ij}^0 & i \neq r, i \neq y, i \neq s \end{cases} = B_2$$

BUT, for  $B_1$  and  $B_2$ ,

at  $i=r$

$$A_{rj}^0 \neq A_{yj}^0$$

and at  $i=s$

$$A_{rj}^0 \neq A_{rj}^0$$

$\therefore [B_1 \neq B_2]$



Hence,

as  $e'$  fails for  $e_3$ ,

there CANNOT be an elementary row operation of the same type as  $e_3$

such that for all cases,

$$e'(e_3(A)) = e_3(e'(A)) = B$$

such that  $A \neq B$ .

Hence, an elementary row operation  $e'$  of the same type as the elementary row operation  $e_i$ ,  $i = \{1, 2, 3\}$

such that

$$e_i(e'(A)) = e'(e_i(A)) = B$$

where  $A \neq B$  is ONLY possible for  $i = 1$ ,

i.e. for  $e_1$ .

It doesn't exist for  $e_2$  and  $e_3$ .



3) To prove:

Every  $m \times n$  matrix ~~is~~ over field  $F$  is row equivalent to a row-reduced matrix.

Proof:

~~To~~ Let  $A_{m \times n}$  be a matrix over field  $F$  such that

$$A_{m \times n} = \begin{bmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{bmatrix}_{m \times n}$$

Now, when elementary row operations,  $e_1, e_2, e_3$  are performed on a matrix,

the resulting matrix is row-equivalent to the original matrix,

where:

$$e_i(A)_{ij} = \begin{cases} cA_{ij} & , i \neq r, c \neq 0 \\ A_{ij} & , i = r \end{cases}$$



$$e_2(A)_{ij}^{00} = \begin{cases} A_{ij}^{00} + c A_{ij}^{00}, & \text{if } i=r \\ A_{ij}^{00}, & i \neq r \end{cases}$$

$$e_3(A)_{ij}^{00} = \begin{cases} A_{ij}^{00}, & i=r \\ A_{ij}^{00}, & i \neq r \\ A_{ij}^{00}, & i \neq r \text{ and } j \neq s \end{cases}$$

Now, for  $A_{m \times n}$ .

row 1:  $\forall A_{i1} = 0 \quad \forall i \in [1, n]$ ,

i.e. row 1 =  $0, 0, 0, \dots, 0$ ,

then it's already in a row reduced format.

However,  $\forall A_{i1} \geq 0 \quad \forall i \in [1, k]$ ,

$$A_{1k_1} \neq 0$$

then, perform,



$$e_1(A)_{ij}^{\circ\circ} \text{ where } r_2 = 1 \text{ and } c = \frac{1}{A_{1k_1}}$$

$$\therefore e_1(A)_{ij}^{\circ\circ} = \begin{cases} \frac{A_{ij}^{\circ\circ}}{A_{1k_1}}, & i = 1 \\ A_{ij}^{\circ\circ}, & i \neq 1 \end{cases}$$

$\therefore$  The leading element of row 1 is now equal to 1

$$\therefore \frac{A_{1k_1}}{A_{1k_1}} = \boxed{1}$$

$$\therefore A \sim \begin{bmatrix} 1 & 2 & 3 & \dots & k_1 & k_2 & n \\ 0 & 0 & 0 & \dots & 1 & \frac{A_{1k_2}}{A_{1k_1}} & \dots \\ \vdots & & & & & & \\ A_{m1} & & & & & & A_{mn} \end{bmatrix}$$

Now, for  $i \neq 1$

$$R_i \leftarrow R_i - (A_{ik_1})(R_1) \rightarrow e_2(A)_{ij}^{\circ\circ}$$

$$\therefore A \sim \begin{bmatrix} 0 & 0 & \dots & 1 & \dots & A_{1n} \\ \vdots & & & 0 & & \\ \vdots & & & 0 & & \\ \vdots & & & 0 & & \end{bmatrix}$$



Similarly,

for row 2.

if  $A_{2i} = 0 \quad \forall i \in [1, n]$ ,

it's already in a row reduced form.

However, if  $A_{2k_2} \neq 0$  for some  $k_2 \in [1, n]$ ,

then:  $r_2 = 2, c = 1$

$$e_1(A_{ij}) = \begin{cases} \frac{A_{2j}^{\infty}}{A_{2k_2}}, & i = 2 \\ A_{ij}^{\infty}, & i \neq 2 \end{cases}$$

$A \sim \begin{bmatrix} 0 & 0 & \dots & 1 & \dots \\ 0 & 0 & \dots & (1) & \dots \\ \vdots & \vdots & & & \end{bmatrix}$

$\rightarrow$  this 1 can be before  $A_{1k_2}$ , after  $A_{1k_2}$ , or in the same line.

NOT on the same line.



Now,

for  $i \neq 2$

$$R_i^0 \leftarrow R_i^0 - (A_{i2})(R_2)$$

$$\therefore A \sim \begin{bmatrix} 0 & 0 & 0 & \dots & 0 & 1 & \dots \\ 0 & 0 & 0 & \dots & 1 & \dots & 0 \\ & & & & & 0 & 0 \\ & & & & & 0 & 0 \\ & & & & & 1 & \\ & & & & & 1 & \end{bmatrix}$$

Now, if we repeat the same steps for all  $n$  rows,

in a finite number of steps (for finite  $n$ ), we will reach a reduced matrix  $R$  such that

$$A \sim R,$$

Hence proved.



4).

II).

$$\begin{cases} x_2 + 2x_3 = 0 \\ x_1 - x_3 = 0 \\ x_1 + x_3 = 0 \end{cases}$$

Now,  $x_1 + x_3 = 0$

and  $x_1 - x_3 = 0$

~~Sub~~ Adding both,

$\therefore 2x_1 = 0$

$x_1 = 0$

(i)

Now,  $x_1 + x_3 = 0$

$\therefore 0 + x_3 = 0$

$x_3 = 0$

(ii)

$x_2 + 2x_3 = 0$

$x_2 + 0 = 0$

$x_2 = 0$

(iii)



Now, if two systems are equivalent, then

Theorem:

Two equivalent systems have the same solution set.

$\therefore$ , as solution set of  $\text{II} = \{(0, 0, 0)\}$ ,

solution set of  $\text{I}$  must be  $\{(0, 0, 0)\}$  for them to be equivalent.

$$\text{I). } \begin{cases} x_1 + 2x_2 + 5x_3 = 0 \\ x_1 + 3x_2 + 8x_3 = 0 \\ -x_1 + x_2 + 3x_3 = 0 \end{cases}$$

Now, 
$$\begin{aligned} x_1 + 3x_2 + 8x_3 &= 0 \\ x_1 + 2x_2 + 5x_3 &= 0 \end{aligned}$$

Subtracting both,

$$x_2 + 3x_3 = 0$$

Let  $x_2 = -3$

$$-3 + 3x_3 = 0$$

$$x_3 = 1$$



Now,  $-x_1 + x_2 + 4x_3 = 0$

$\therefore -x_1 + (-3) + 4 = 0$

$-x_1 + 1 = 0$

$x_1 = 1$

Now,  $X = (1, -3, 1)$

Putting in all equations

i).  $x_1 + 2x_2 + 5x_3 = 1 - 6 + 5 = 0$

ii).  $x_1 + 3x_2 + 8x_3 = 1 - 9 + 8 = 0$

iii).  $-x_1 + x_2 + 4x_3 = -1 - 3 + 4 = 0$

Hence,  $I$  has  $X = (1, -3, 1)$  in its solution set as well.

However,  $X \in I$  but  $X \notin II$

where  $I$  and  $II$  are solution sets of  $I$  and  $II$ .

Hence,  $I \neq II$ . Thus, as the set of equations have DIFFERENT solutions.



sets, they are NOT ~~equal~~ equivalent systems.



Method 2 :

II).

Augmented Matrix =

$$\left[ \begin{array}{ccc|c} 0 & 1 & 2 & 1 \\ 1 & 0 & -1 & 0 \\ 1 & 0 & 1 & 0 \end{array} \right]$$

Swap  $R_1$  and  $R_3$

$$\left[ \begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$R_2 \leftarrow R_2 - R_1$$

$$\left[ \begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & 0 \\ 0 & 1 & 2 & 1 \end{array} \right]$$

$$R_2 \leftarrow R_2 + R_3$$



$$\left[ \begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 \end{array} \right]$$

$$R_3 \leftarrow \frac{R_3}{2} - \frac{R_2}{2}$$

$$\left[ \begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$R_1 \leftarrow R_1 - R_3$$

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\therefore n = 0$$

$$n_2 = 20$$

$$n_3 = 20,$$

for I).

Augmented Matrix:

$$\left[ \begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 1 & 3 & 8 & 0 \\ -1 & 1 & 4 & 0 \end{array} \right]$$

$$R_2 \leftarrow R_2 - R_1 \quad R_3 \leftarrow R_3 + R_1$$



$$\left[ \begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 3 & 9 & 0 \end{array} \right]$$

$$R_3 \leftarrow R_3 - 3R_2$$

$$\left[ \begin{array}{ccc|c} 1 & 2 & 5 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$\therefore$  for  $x_2 = -3,$

$$(x_1, x_2, x_3) = (1, -3, 1)$$